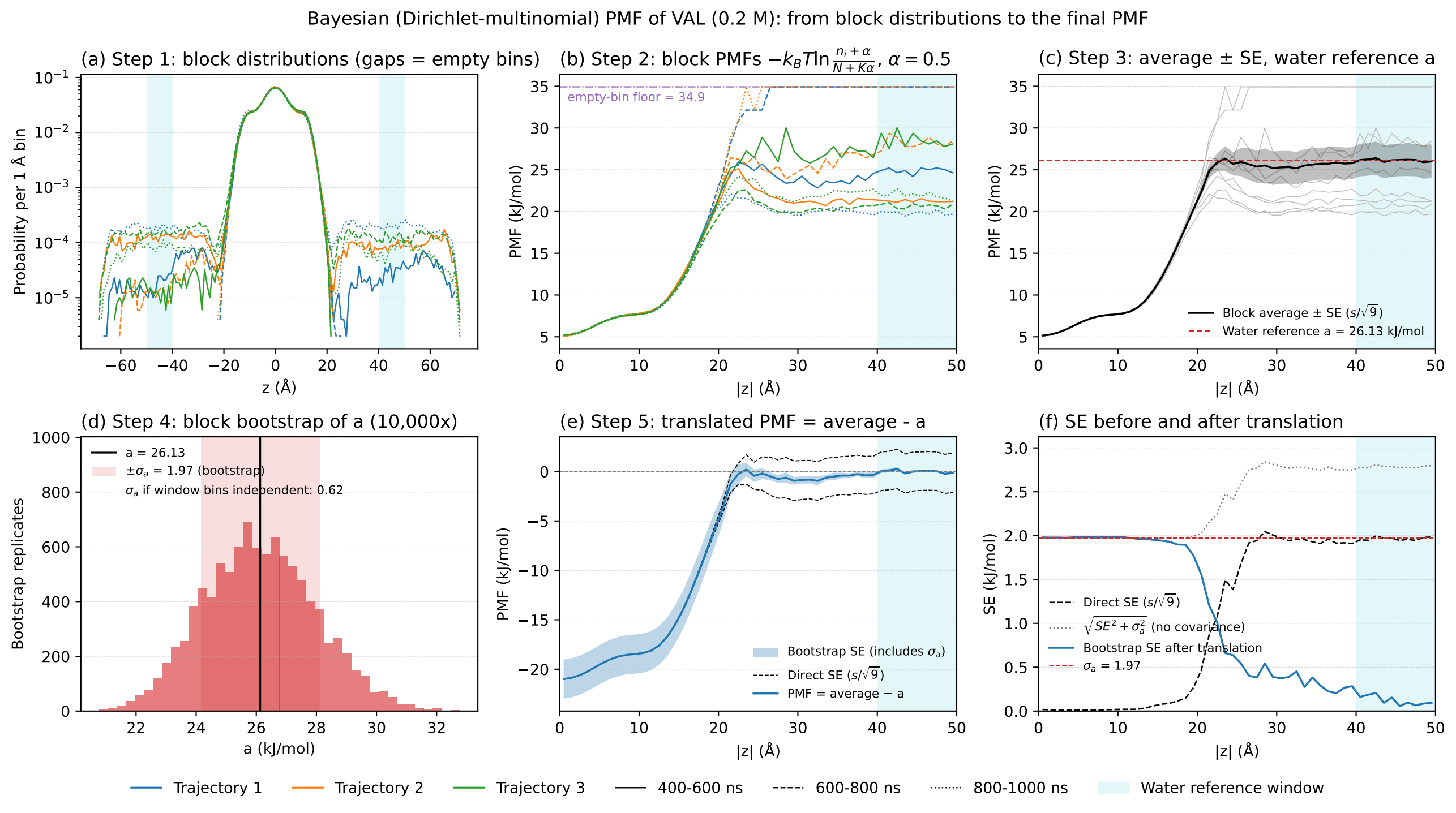
Bayesian (Dirichlet-multinomial) PMF of VAL (0.2 M): from block distributions to the final PMF
(a) Step 1: block distributions (gaps = empty bins)
Probability per 1 Å bin
z (Å)
(b) Step 2: block PMFs $-k_BT\ln\frac{n_i+\alpha}{N+K\alpha}$, $\alpha = 0.5$
empty-bin floor = 34.9
PMF (kJ/mol)
|z| (Å)
(c) Step 3: average ± SE, water reference a
Block average ± SE ($s/\sqrt{9}$)
Water reference a = 26.13 kJ/mol
PMF (kJ/mol)
|z| (Å)
(d) Step 4: block bootstrap of a (10,000x)
a = 26.13
$\pm\sigma_a$ = 1.97 (bootstrap)
σ_a if window bins independent: 0.62
Bootstrap replicates
a (kJ/mol)
(e) Step 5: translated PMF = average - a
Bootstrap SE (includes σ_a)
Direct SE ($s/\sqrt{9}$)
PMF = average − a
PMF (kJ/mol)
|z| (Å)
(f) SE before and after translation
Direct SE ($s/\sqrt{9}$)
$\sqrt{SE^2+\sigma_a^2}$ (no covariance)
Bootstrap SE after translation
σ_a = 1.97
SE (kJ/mol)
|z| (Å)
Trajectory 1
Trajectory 2
Trajectory 3
400-600 ns
600-800 ns
800-1000 ns
Water reference window